

BEN HORWITZ

benhorwitz05@gmail.com

ben-horwitz.com

EDUCATION	DARTMOUTH COLLEGE Candidate for B.E. in Mechanical Engineering GPA: 3.96	Hanover, NH Expected May 2028
	MIDDLEBURY COLLEGE Candidate for B.A. in Mathematics	Middlebury, VT Expected May 2027
WORK EXPERIENCE	DESIGN AND MANUFACTURING LAB, GEORGIA TECH Additive Manufacturing Engineering Intern - Led the development of a novel SLA printer that prints high-viscosity silicones. - Held planar tolerances of 0.001" across an 18" build plate. - Built PLC controls and designed custom hardware and software. - Developed a custom slicer that outputs on 8K LCD screen.	Atlanta, GA Summer 2026
	CENTER FOR ADDITIVE MANUFACTURING, SLU Additive Manufacturing Engineering Intern - Excelled at printing on FDM, SLS, SLA, and PolyJet technologies. - Designed a custom reaction chamber for an organic chemistry lab. - Helped lead outreach events at a Technical College and Science Center.	St. Louis, MO Summer 2025
	MECHANICAL ADVANTAGE LLC. Engineering Intern - Accelerated workflow by manufacturing 10 quick-release harperizer lids. - Programmed an automated "bar-pull" to decrease manual labor. - Collaborated with three other interns and 10 full-time professionals.	Middlebury, VT Summer 2024
	TISSUE ENGINEERING SCAFFOLD FABRICATION LABORATORY, ST. LOUIS UNIVERSITY Research Intern - Collaborated with graduate researchers to modify a 3D printer for synthetic matrix fabrication. - Wrote a board-reviewed research paper and was awarded "Best Presentation".	St. Louis, MO Summer 2022
	MIDDLEBURY COLLEGE MAKERSPACE Teaching Assistant - Supervised and trained over 40 community members on equipment. - Programmed and manufactured a golf putter on a 4-axis mill. - Designed an escapement mechanism to be used in a class setting.	Middlebury, VT Fall 2024 - Spring 2025
	MIDDLEBURY ENGINEERING SOCIETY President - Spearheaded the re-branding and organization of the club. - Developed and led seminars on 3D printing and laser cutting for 15+ members. - Created long-standing relationship between club and makerspace.	Middlebury, VT January 2024 - Present
PROJECTS	HIGH-VISCOSITY VAT PHOTOPOLYMERIZATION PRINTER - Designed mechanical and PLC control systems for a custom additive manufacturing platform - Enabled the creation of silicone actuators for eventual use in soft robotics. 4TH AXIS CNC PUTTER - Designed and manufactured a custom 303 stainless steel putter.	
TECHNICAL SKILLS	CNC machinery (HAAS mill 3rd and 4th axis, manual lathe) Additive manufacturing (SLA, SLS, FDM, RFAM) CAD, CAM, FEA (SolidWorks, Autodesk Fusion 360, Onshape) Coding (Matlab, Python, experience in R) Controls (Beckhoff PLC, Indradrive)	

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	TISSUE ENGINEERING SCAFFOLD FABRICATION LABORATORY, ST. LOUIS UNIVERSITY Research Intern • Worked alongside graduate students for 20 hours a week. • Modified a 3D printer to create synthetic matrices, where findings were shared in a peer-reviewed research paper and presentation.	St. Louis, MO Summer 2022
	MIDDLEBURY ENGINEERING SOCIETY Co-President • Spearheaded the re-branding and organization of the club. • Developed and led seminars on 3D printing and laser cutting for 15+ members. • Created long-standing relationship between club and makerspace.	Middlebury, VT January 2024 - Present
	MIDDLEBURY ADMISSIONS OFFICE Tour Guide • Led over 20 tours for prospective students to showcase the college. • Guided 400+ students and family members.	Middlebury, VT February 2024 - Present
CAMPUS/ COMMUNITY ACTIVITIES	PLANTING SHADE President • Led 501(c)(3) to plant over 15,000 trees in several cities across the US. • Interviewed chapter leader candidates to expand the non-profit to 11 chapters worldwide.	St. Louis, MO January 2021 - May 2023
	SKILLS • CNC machinery (HAAS mill 3rd and 4th axis, manual lathe) • Additive manufacturing (Bambu, Stratasys, Formlabs) • CAD, CAM, FEA (SolidWorks, Autodesk Fusion 360) • Coding (Matlab, Python, experience in R) • Laser cutting (Epilog) • Graphic design (Canva, Adobe Illustrator)	

